AP Calculus BC Unit 8 Review Book
Applications of Integration

A complete English review guide for final AP review, written to be beginner-friendly, detailed, and exam-focused.

Goal of this unit. Unit 8 asks you to use integrals in meaningful geometric and contextual ways. You should be able to recognize whether a question is about average value, net change, total distance, area between curves, volume, or arc length, and then set up the correct integral with the correct variable, radii, limits, and interpretation.

Chapter	Main idea
8.1	Average value of a function on an interval
8.2	Using integrals to connect position, velocity, and acceleration
8.3	Accumulation functions and definite integrals in applied contexts
8.4–8.6	Area between curves
8.7–8.8	Volumes from cross sections
8.9–8.12	Disc and washer methods of revolution
8.13	Arc length and total distance

Core Formula Sheet

Start with these formulas. The rest of the book explains how and when to use them.

$$f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

$$s(b) - s(a) = \int_a^b v(t)\,dt, \qquad v(b) - v(a) = \int_a^b a(t)\,dt$$

$$\text{Total distance} = \int_a^b |v(t)|\,dt$$

$$A = \int_a^b (\text{top} - \text{bottom})\,dx, \qquad A = \int_c^d (\text{right} - \text{left})\,dy$$

$$V = \int A(\text{cross section})\,d(\text{variable})$$

$$V = \pi \int R^2\,d(\text{variable}), \qquad V = \pi \int (R^2 - r^2)\,d(\text{variable})$$

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx$$

8.1 Finding the Average Value of a Function on an Interval

The average value of a function is the average height of its graph over an interval. It is *not* the same thing as average rate of change. Average rate of change uses a slope formula; average value uses signed area divided by interval length.

$$f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

- Use the same units as the original function, not units per input unit.
- Think of a horizontal line with height f_{avg} that would produce the same signed area on $[a, b]$.
- The average value can be negative if the function is mostly below the x-axis.

Common AP mistakes.

- Using $\dfrac{f(b)-f(a)}{b-a}$, which is average rate of change, not average value.
- Forgetting the factor $\dfrac{1}{b-a}$.

Worked example. Find the average value of $f(x) = x^2$ on $[0, 2]$.

$$f_{\text{avg}} = \frac{1}{2-0}\int_0^2 x^2\,dx = \frac{1}{2}\left[\frac{x^3}{3}\right]_0^2 = \frac{1}{2}\cdot\frac{8}{3} = \frac{4}{3}$$

So the average value is $\frac{4}{3}$.

8.2 Connecting Position, Velocity, and Acceleration Functions Using Integrals

In motion problems, differentiation moves downward from position to velocity to acceleration, and integration reverses the process. Definite integrals give net change.

$$v(t) = s'(t), \qquad a(t) = v'(t) = s''(t)$$

$$s(b) - s(a) = \int_a^b v(t)\,dt$$

$$v(b) - v(a) = \int_a^b a(t)\,dt$$

$$\text{Total distance} = \int_a^b |v(t)|\,dt$$

- Displacement is signed. Negative velocity contributes negative displacement.
- Total distance ignores direction, so it uses absolute value.
- If you know acceleration and one velocity value, integrate acceleration and use the initial condition.
- If you know velocity and one position value, integrate velocity and use the initial condition.

Common AP mistakes.

- Confusing displacement and total distance.
- Forgetting to find where $v(t) = 0$ before setting up total distance.

Worked example. Let $v(t) = t^2 - 4t + 3$ on $0 \le t \le 4$. Find the displacement and explain how total distance is set up.

$$\int_0^4 (t^2 - 4t + 3)\,dt = \left[\frac{t^3}{3} - 2t^2 + 3t\right]_0^4 = -\frac{4}{3}$$

So the displacement is $-\frac{4}{3}$. For total distance, solve

$$t^2 - 4t + 3 = (t-1)(t-3) = 0 \Rightarrow t = 1, 3$$

Then split the absolute-value integral over $[0, 1]$, $[1, 3]$, and $[3, 4]$.

8.3 Using Accumulation Functions and Definite Integrals in Applied Contexts

An accumulation function tracks how a quantity changes from a starting amount. If $R(t)$ is a rate, then integrating R gives total accumulated change.

$$Q(x) = Q(a) + \int_a^x R(t)\,dt$$

- Separate the initial amount from the accumulated change.
- A rate in units per minute becomes a total amount in units after integration.
- A negative rate means the quantity is decreasing.

Worked example. A tank starts with 120 gallons. Water enters at rate $R(t) = 5 + \sin t$ gallons per minute. Write the amount of water after t minutes.

$$W(t) = 120 + \int_0^t (5 + \sin u)\,du$$

The constant 120 is the initial amount. The integral gives the net change from time 0 to time t.

8.4 Finding the Area Between Curves Expressed as Functions of x

With vertical slices, area between curves is found by subtracting bottom from top. This is the standard x-based setup.

$$A = \int_a^b (f(x) - g(x))\,dx \quad \text{if } f(x) \ge g(x)$$

- Solve for intersection points first.

- Check which graph is above on the interval.
- If the upper graph changes, split the integral.

Worked example. Find the area enclosed by $y = x$ and $y = x^2$.

$$x = x^2 \Rightarrow x = 0, 1$$

On $0 \le x \le 1$, the line $y = x$ is above the parabola $y = x^2$.

$$A = \int_0^1 (x - x^2)\,dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{1}{6}$$

8.5 Finding the Area Between Curves Expressed as Functions of y

With horizontal slices, subtract left from right and integrate with respect to y. This is often better for sideways parabolas or regions that would require splitting if you used dx.

$$A = \int_c^d (x_{\text{right}} - x_{\text{left}})\,dy$$

- Rewrite each boundary curve as $x =$ a function of y.
- Use y-values as limits, not x-values.
- This method is useful whenever horizontal slices cut the region cleanly.

Worked example. For the region bounded by $x = y^2$ and $x = y + 2$, the setup is

$$A = \int_{y_1}^{y_2} ((y + 2) - y^2)\,dy$$

where y_1 and y_2 come from solving $y^2 = y + 2$.

8.6 Finding the Area Between Curves That Intersect at More Than Two Points

When curves meet at several points, the top-minus-bottom or right-minus-left relationship can change. Then the total area must be written as a sum of several integrals.

$$A = \int_a^b (\text{top} - \text{bottom})\,dx + \int_b^c (\text{top} - \text{bottom})\,dx + \cdots$$

- Find all relevant intersection points.
- Use a sketch or sign chart to determine which curve is on top on each interval.
- Add the positive areas. Do not allow cancellation.

Worked example. If two graphs cross at $x = a$, $x = b$, and $x = c$, and the top graph changes at $x = b$, then you must split the integral at $x = b$. On the AP exam, the setup is often the main idea being tested.

8.7 Volumes with Cross Sections: Squares and Rectangles

For solids with known cross sections, the volume is the integral of the cross-sectional area. First find the base of a typical slice, then turn that base into an area formula.

$$V = \int_a^b A(x)\,dx \qquad \text{or} \qquad V = \int_c^d A(y)\,dy$$

- For a square, area is side squared.
- For a rectangle, area is length times width, and one dimension is often given as a multiple of the base.
- The base of the slice is usually the difference of two curves.

Worked example. If cross sections perpendicular to the x-axis are squares and the base runs from $y = f(x)$ down to $y = g(x)$, then

$$\text{side} = f(x) - g(x), \qquad A(x) = \big(f(x) - g(x)\big)^2$$

$$V = \int_a^b \big(f(x) - g(x)\big)^2\,dx$$

8.8 Volumes with Cross Sections: Triangles and Semicircles

The cross-section method stays the same, but the area formula depends on the shape. The most common AP special cases are equilateral triangles and semicircles.

$$A_{\text{equilateral}} = \frac{\sqrt{3}}{4}s^2$$

$$A_{\text{semicircle}} = \frac{1}{2}\pi r^2 = \frac{\pi}{8}d^2$$

- For a semicircle, check whether the problem gives the radius or the diameter.
- For triangle cross sections, use exactly the triangle information stated in the problem.

Worked example. If each cross section is a semicircle with diameter $f(x) - g(x)$, then

$$A(x) = \frac{\pi}{8}\big(f(x) - g(x)\big)^2$$

$$V = \int_a^b \frac{\pi}{8}\big(f(x) - g(x)\big)^2\,dx$$

8.9 Volume with the Disc Method: Revolving Around the x-axis or y-axis

The disc method applies when a region rotates into a solid with no hole. Each slice becomes a solid disc, so the volume formula uses one radius only.

$$V = \pi \int_a^b [R(x)]^2 \, dx$$
$$V = \pi \int_c^d [R(y)]^2 \, dy$$

- The radius is always the distance from the curve to the axis of rotation.
- If rotating around the x-axis with vertical slices, the radius is a vertical distance.
- If rotating around the y-axis with horizontal slices, the radius is a horizontal distance.

Worked example. Rotate the region under $y = \sqrt{x}$ from $x = 0$ to $x = 4$ about the x-axis.

$$V = \pi \int_0^4 (\sqrt{x})^2 \, dx = \pi \int_0^4 x \, dx = 8\pi$$

8.10 Volume with the Disc Method: Revolving Around Other Axes

If the axis of rotation is $y = k$ or $x = h$ instead of one of the coordinate axes, the disc method still works. The radius must now be written as a distance to the shifted axis.

$$V = \pi \int [\text{distance to the axis}]^2 \, d(\text{variable})$$

- Around $y = k$, use a vertical distance such as $|f(x) - k|$.
- Around $x = h$, use a horizontal distance such as $|g(y) - h|$.

Worked example. If a region under $y = f(x)$ is revolved about $y = 5$ and reaches the axis with no hole, then

$$R(x) = |5 - f(x)|, \qquad V = \pi \int_a^b (5 - f(x))^2 \, dx$$

8.11 Volume with the Washer Method: Revolving Around the x-axis or y-axis

A washer is a disc with a hole. When the rotated region does not touch the axis of rotation, every slice has an outer radius and an inner radius.

$$V = \pi \int (R^2 - r^2) \, d(\text{variable})$$

- R is the outer radius, farther from the axis.

- r is the inner radius, nearer to the axis.
- Square both radii before subtracting.

Worked example. If $f(x)$ is above $g(x)$ and both are above the x-axis, then revolving the region between them about the x-axis gives

$$V = \pi \int_a^b \left([f(x)]^2 - [g(x)]^2 \right) \, dx$$

8.12 Volume with the Washer Method: Revolving Around Other Axes

Shifted-axis washer problems combine two ideas: use distances for radii, and subtract inner from outer.

$$V = \pi \int_a^b \left(R(x)^2 - r(x)^2 \right) \, dx$$

- For rotation around $y = k$, both radii are vertical distances to $y = k$.
- For rotation around $x = h$, both radii are horizontal distances to $x = h$.
- A quick sketch prevents almost every setup mistake.

Worked example. If the region between $y = f(x)$ and $y = g(x)$ is revolved around $y = 2$, then

$$R(x) = \text{farther distance to } y = 2, \qquad r(x) = \text{nearer distance to } y = 2$$

Use the sketch to decide which expression is outer and which is inner.

8.13 (BC Only) The Arc Length of a Smooth Planar Curve and Distance Traveled

Arc length measures the length of a graph. Total distance measures how far a moving particle travels. Both involve accumulating tiny lengths, but they use different formulas.

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} \, dx$$

$$\text{Total distance} = \int_a^b |v(t)| \, dt$$

- Use the arc-length formula only for a graph $y = f(x)$.
- Use total distance in one-dimensional motion problems.
- Many AP questions only ask for the correct setup, especially when the integral is difficult.

Worked example. Find the arc length of $y = x^2$ on $[0, 1]$.

$$y' = 2x$$

$$L = \int_0^1 \sqrt{1 + (2x)^2}\, dx = \int_0^1 \sqrt{1 + 4x^2}\, dx$$

How to Choose the Correct Unit 8 Method

Fast decision guide.

- **Average value?** Use one over interval length times a definite integral.
- **Motion?** Integrate acceleration to get velocity, velocity to get position, and use absolute value for total distance.
- **Area between curves?** Decide between vertical slices (top minus bottom) and horizontal slices (right minus left).
- **Cross sections?** Volume is always the integral of the cross-sectional area.
- **No hole after rotation?** Disc method.
- **Hole after rotation?** Washer method.
- **Axis shifted?** Radius means distance to the axis, not just the function value.
- **Curve length?** Arc length formula.

Most Important AP Mistakes to Avoid

- Do not mix up average value and average rate of change.
- Do not use a plain velocity integral for total distance when velocity changes sign.
- Always solve for intersection points before setting up area-between-curves integrals.
- For dy area problems, use right minus left.
- For solids of revolution, the radius is always a distance to the axis.
- Disc and washer formulas require squared radii.
- For cross sections, integrate slice area, not slice length.
- For arc length, keep the square root.

Final exam advice. On free-response questions, a correct setup with clear notation often earns most of the available credit. Label radii, identify top and bottom or right and left explicitly, and include units whenever the question is in context.